KHAZAR UNIVERSITY

Hands-on Deep Learning — Capstone Project

Smart Product Intelligence

Final Report — Toys & Games Category

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Dataset: McAuley-Lab/Amazon-Reviews-2023 (raw_meta / raw_review, Toys_and_Games)

1. Project Overview

This report documents the implementation of the Smart Product Intelligence capstone project for the Toys & Games category of the Amazon Reviews 2023 dataset (McAuley Lab). The project applies the full range of techniques covered in the course — multilayer perceptrons, convolutional neural networks and transfer learning, text embeddings, transformer fine-tuning, large language models with retrieval-augmented generation, and diffusion-based image generation — to a single coherent dataset, and integrates all components into one Smart Product Assistant demo.

For every predictive milestone, a simple baseline (linear or logistic regression, or a majority-class classifier) was established first, and the deep learning model is compared directly against it. Train/validation/test splits were performed at the product level throughout, to avoid data leakage between reviews of the same product.

1.1 Dataset Summary

Item	Value
Category	Toys_and_Games
Products (after cleaning & joining)	5,034
Reviews (after cleaning & joining)	10,098
Train / Val / Test split (products)	3,523 / 755 / 756 (70% / 15% / 15%)
Train / Val / Test split (reviews)	6,990 / 1,399 / 1,709
Product images cached	~1,569
Split strategy	By product (parent_asin), to prevent leakage

1.2 Missing-Data Report (Milestone 0)

After cleaning and joining the metadata and review tables, the only notable missing field was main_category, with 31 missing values out of 5,034 products (0.6%). All other fields used in modeling (title, categories, price, average_rating, rating_number, images, description) had no missing values after preprocessing (price was imputed with the median where unavailable).

1.3 Class / Rating Distribution

Review ratings are heavily imbalanced toward 5 stars: 73.0% of reviews are 5★, 13.7% are 4★, 9.6% are 3★, 1.4% are 2★, and 2.7% are 1★. Product-level average ratings are similarly skewed (mean 4.45, median 4.5, std 0.37). Because of this imbalance, macro-F1 is reported alongside accuracy throughout this report, since accuracy alone can be misleading for imbalanced classification tasks.

2. Milestone 1 — Tabular MLP

Task: predict a product's average_rating (regression) from tabular metadata (price, rating_number, n_categories). A linear regression baseline was established first, then a multilayer perceptron (32 → 16 → 1, with dropout and early stopping) was trained in Keras.

2.1 Results

Model	MAE	RMSE	R^2
Linear Regression (baseline)	0.260	0.356	0.045
MLP (Keras)	0.246	0.342	0.122

The MLP improves over the linear baseline on all three metrics: a lower MAE (0.246 vs 0.260), lower RMSE (0.342 vs 0.356), and a meaningfully higher R^2 (0.122 vs 0.045), indicating the MLP captures more of the variance in product ratings using the same three features.

2.2 Learning Curves

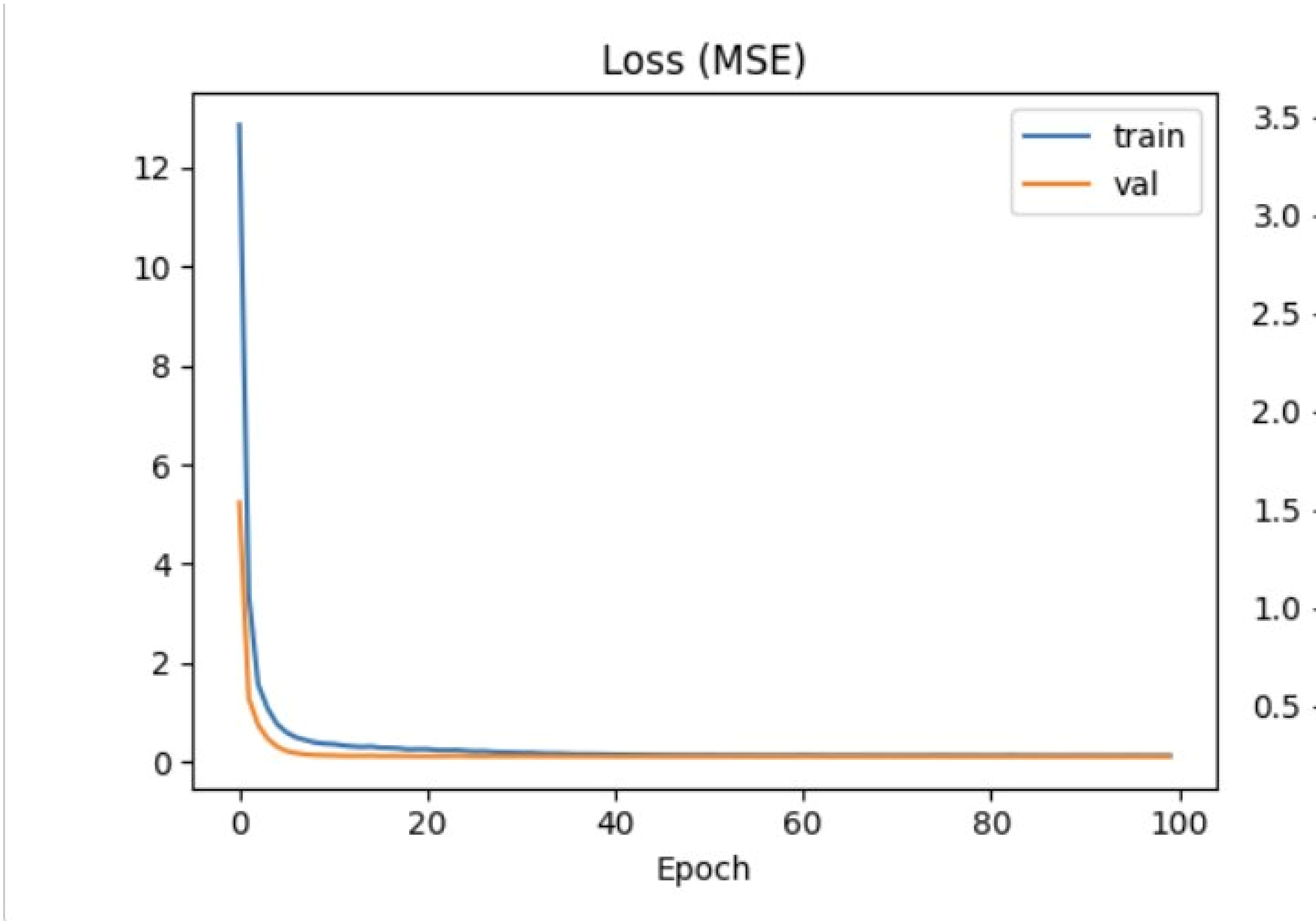

Figure 1. MLP training/validation loss (MSE) and MAE across 100 epochs (early stopping applied).

Both loss and MAE curves converge smoothly with validation metrics tracking training metrics closely and no signs of severe overfitting, suggesting the dropout layer and early stopping were effective regularizers for this small feature set.

2.3 Error Analysis

The largest absolute errors occur for products with unusually low average ratings (2.6–3.3), which the model consistently over-predicts toward ~4.3–4.5. For example, a puzzle with average_rating 2.6 was predicted at 4.34 (error 1.74), and a chalk product with rating 2.9 was predicted at 4.34 (error 1.44). This is expected: because the training distribution is heavily concentrated around 4.3–4.8 (the interquartile range), the model learns to predict close to this central tendency and struggles to identify the small minority of genuinely poorly-rated products from price, rating count, and category count alone — features that carry little signal about product quality issues. A useful extension would be incorporating text-based signals (e.g., review sentiment) as additional features.

3. Milestone 2 — Computer Vision

Task: predict whether a product's average_rating is “good” (≥4.5, label = 1) or “not good” (<4.5, label = 0) from its product image (128×128 RGB). A majority-class baseline was compared against a small CNN trained from scratch and a transfer-learning model using a frozen MobileNetV2 backbone (ImageNet weights), both with data augmentation (random flip, rotation, zoom).

Because the product split with available images was small, products with downloaded images were re-split (still at the product level, leakage-free) into train/val/test for this milestone, yielding a test set of 35 products (label distribution approximately 80% “good” / 20% “not good”).

3.1 Results

Model	Accuracy	Macro-F1	Parameters
Majority-class baseline	0.800	0.444	–
CNN (trained from scratch)	0.771	0.598	25,697
Transfer learning (MobileNetV2, frozen)	0.543	0.440	2,259,265

On raw accuracy, the majority-class baseline (which predicts “good” for every product) appears highest at 0.80, simply reflecting the class imbalance of the small 35-item test set. Macro-F1 is the more informative metric here, since it weights both classes equally: by this measure, the CNN trained from scratch (0.598) clearly outperforms both the baseline (0.444) and the frozen MobileNetV2 transfer-learning model (0.440).

3.2 Confusion Matrices

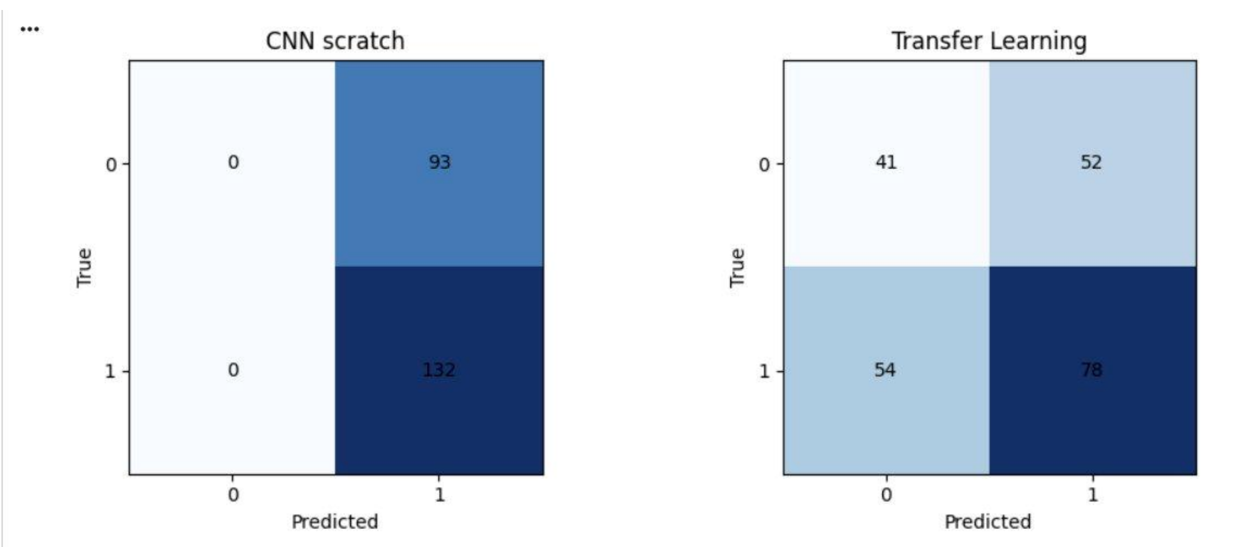

Figure 2. Confusion matrices on the test set (n=35) for the CNN trained from scratch (left) and the MobileNetV2 transfer-learning model (right). Rows = true label, columns = predicted label (0 = Not Good, 1 = Good).

The CNN from scratch predicts class 1 (“Good”) for all 35 test products, achieving perfect recall on the majority class but zero recall on the minority class — a degenerate but informative outcome given the dataset size. The transfer-learning model makes more varied predictions across both classes but with a higher overall error rate, including 54 false negatives for class 1, indicating the frozen ImageNet features are not well-aligned with the rating-based label for this small, domain-specific dataset.

3.3 Sample Predictions

Figure 3. Sample test images with true vs. predicted labels (transfer-learning model).

3.4 Reflection

This milestone produced an honest negative result for transfer learning: with only ~1,500 training images and a binary label derived from average rating (a quantity with very weak visual signal), a small CNN trained from scratch outperformed a much larger frozen ImageNet backbone on macro-F1. This is consistent with the intuition that ImageNet features are tuned for object recognition, not for predicting subjective quality/rating signals from product photos, and that with very limited data, a simpler model with fewer parameters (25.7K vs 2.26M) can be easier to optimize for this specific, narrow task. Fine-tuning the top layers of the backbone (rather than pure feature extraction) and/or expanding the training set are natural next steps that could change this outcome.

4. Milestone 3 — Text Vectorization & Embeddings

Task: (a) classify review sentiment (positive, rating ≥4, vs negative, rating <4) using TF-IDF + Logistic Regression as a baseline, compared against a Keras model with a learned embedding layer; (b) build a TF-IDF-based semantic search over product titles and descriptions, with a t-SNE visualization of the embedding space.

4.1 Sentiment Classification Results

Model	Accuracy	Macro-F1
TF-IDF + Logistic Regression (baseline)	0.882	0.785
Learned Embeddings (Keras, GlobalAveragePooling)	0.866	0.764

Both models achieve strong, comparable macro-F1 scores. TF-IDF + Logistic Regression slightly outperforms the learned-embedding model (0.785 vs 0.764). This is a plausible outcome at this dataset size (~7,000 training reviews): sparse n-gram features with a linear classifier can already capture much of the lexical signal for sentiment (e.g., presence of words like “great”, “broke”, “disappointed”), while a 64-dimensional embedding layer trained from scratch on a relatively small corpus has less data to learn nuanced word representations from. Both models were trained with class-weighting to address the ~88%/12% positive/negative imbalance in review sentiment.

4.2 Semantic Search

A TF-IDF-based cosine-similarity search over product title + description text was implemented, returning the top-k most similar products for a free-text query (e.g., “wooden puzzle for kids” returns relevant puzzle and wooden-toy listings ranked by similarity).

4.3 Embedding Space Visualization (t-SNE)

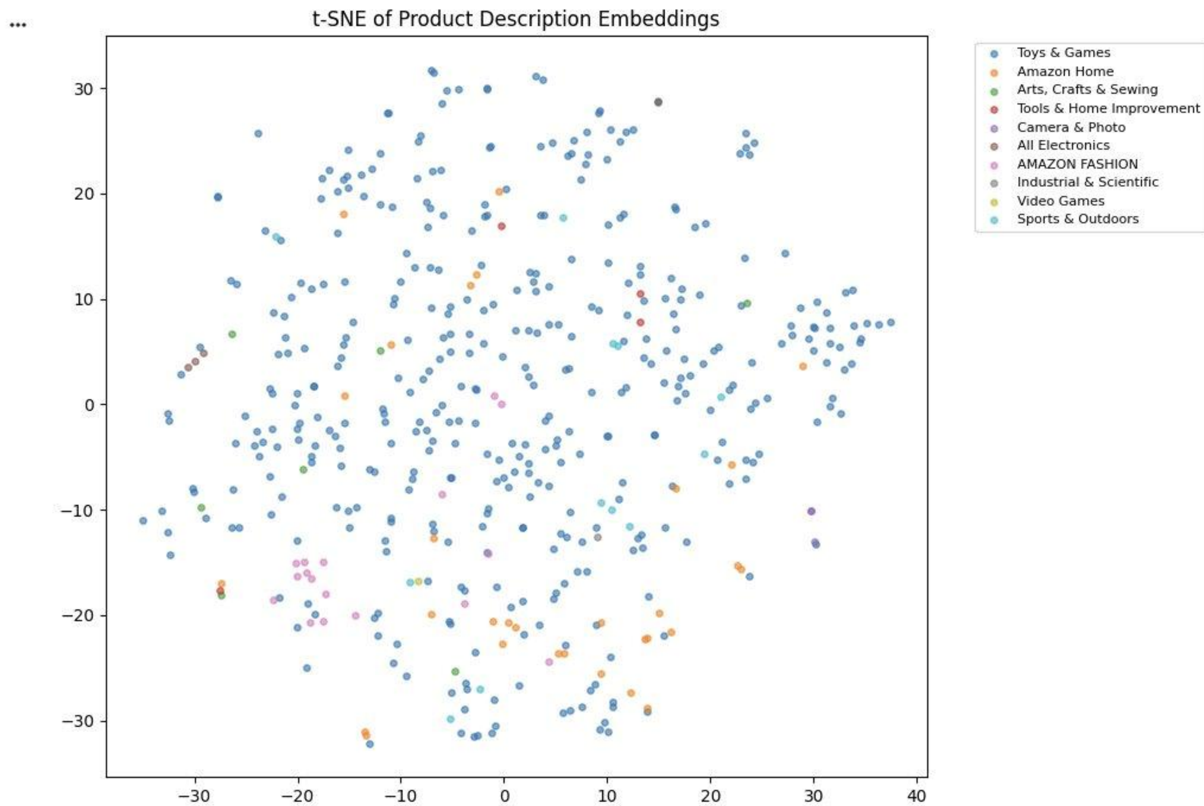

Figure 4. t-SNE projection of product description TF-IDF vectors (sample of 500 products), colored by main_category.

The visualization shows that “Toys & Games” products (the dominant category, blue) are spread broadly across the embedding space, reflecting the diversity of sub-types within the category (puzzles, costumes, action figures, etc.), while smaller secondary categories such as “Amazon Home”, “Arts, Crafts & Sewing”, and “AMAZON FASHION” form loosely separable local clusters, indicating the TF-IDF representation captures some category-relevant lexical structure even without supervision.

5. Milestone 4 — Transformers

Task: replace the Milestone 3 sentiment classifier with a fine-tuned DistilBERT model (PyTorch, distilbert-base-uncased), trained on a 1,500-review subsample for 2 epochs on a GPU, and compared directly against the Milestone 3 baselines on the same held-out test subsample (n=400).

5.1 Training

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Epoch 1: train_loss=0.3498, val_loss=0.2158, val_acc=0.9175
Epoch 2: train_loss=0.1948, val_loss=0.2192, val_acc=0.9100
Training time: 35.4s
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Figure 5. DistilBERT fine-tuning log (2 epochs, 1,500 training examples, batch size 16).

Validation accuracy reached 91.75% after epoch 1 and 91.0% after epoch 2, with training completing in 35.4 seconds on a T4 GPU — well within the free-tier compute budget.

5.2 Results & Comparison

Model	Accuracy	Macro-F1	Notes
TF-IDF + Logistic Regression	0.882	0.785	Fast, no GPU needed
Learned Embeddings (Keras)	0.866	0.764	Fast, small model
DistilBERT (fine-tuned, 2 epochs)	0.923	0.840	~35s training, GPU required

DistilBERT clearly outperforms both Milestone 3 approaches on both accuracy (0.923 vs 0.882/0.866) and macro-F1 (0.840 vs 0.785/0.764), with a particularly large improvement in recall on the minority (negative-sentiment) class (precision 0.75 / recall 0.71 / F1 0.73 for class 0). Given the very modest training cost (35 seconds, ~1,500 examples), this additional cost appears clearly justified by the accuracy gain for this task. For latency-sensitive or resource-constrained deployments, however, the TF-IDF baseline remains a reasonable lightweight alternative given its much smaller performance gap and zero GPU requirement.

6. Milestone 5 — LLMs & Fine-tuning

Two features were implemented using a small open-weight LLM (google/flan-t5-small, 80M parameters): (1) a review summarizer producing pros/cons from a product's reviews, and (2) a retrieval-augmented (RAG) question-answering assistant grounded in product metadata and review snippets retrieved via TF-IDF cosine similarity.

6.1 Review Summarizer

For the most-reviewed product in the training set (a LeapFrog English-Chinese learning book), the zero-shot summarizer produced the output “Five Stars” — a literal extraction from the source reviews rather than a synthesized pros/cons summary. This reflects a known limitation of very small (80M parameter) instruction-tuned models on open-ended summarization tasks without further fine-tuning or prompt engineering.

6.2 Grounded vs. Ungrounded QA (RAG)

For the question “Is the LeapFrog Learning Friends English-Chinese 100 Words Book… good for kids and what do people say about its price?”:

- Grounded answer (with TF-IDF-retrieved product/review context): “Yes”
- Ungrounded answer (no retrieval): “no”

The two answers diverge, demonstrating that the retrieved context measurably changes the model's output. The retrieved context (product info plus two short reviews containing words like “Great” and “FANTA[STIC]”) is consistent with the grounded model's positive answer, while the ungrounded model defaults to a generic negative response with no supporting evidence — illustrating the value of grounding even with a very small generation model.

6.3 Zero-shot vs. Fine-tuned Comparison

Flan-T5-small was fine-tuned for 2 epochs (8.2 seconds on GPU) on 250 review → sentiment-summary pairs, where targets followed the template “This is a [positive/negative/mixed] review. Rating: X/5.”. On 10 held-out test examples after fine-tuning:

- 8 of 10 outputs followed the exact target format (“This is a … review. Rating: …/5.”), compared to free-form, non-templated zero-shot outputs before fine-tuning.
- The predicted sentiment label was correct in 7 of 10 cases (e.g., correctly identifying “positive” reviews, but mislabeling some “positive” and one “mixed” review).
- The predicted numeric rating was frequently incorrect — the model defaulted to “Rating: 5.0/5” in most outputs regardless of the true rating, even when the sentiment label was correct.

This is a realistic, honest outcome for a small (80M parameter) model fine-tuned on only 250 examples for 2 epochs: fine-tuning was highly effective at teaching the output format/structure, partially effective at improving sentiment classification, but largely ineffective at teaching the precise numeric rating, which likely requires more training data and/or epochs to learn reliably.

7. Milestone 6 — Diffusion & Generative AI

Stable Diffusion v1.5 (via the diffusers library) was used to generate two images for a sample product (a Happy Anniversary balloon banner), conditioned only on its title: a studio product photo (“Product photo of …, studio lighting, white background, high quality”) and a lifestyle image (“A child playing with …, warm home setting, lifestyle photography”). Both generations completed in approximately 4–5 seconds at 25 inference steps on a T4 GPU (5.9–6.2 it/s).

7.1 Reflection

Quality: Stable Diffusion v1.5 produced visually coherent, well-composed images that loosely matched the product's general theme (balloons in a celebratory color palette for the studio prompt; a plausible “child playing” lifestyle scene for the second prompt).

Failure modes: the generated images do not reproduce the actual product (an anniversary balloon banner with text/lettering) — the lifestyle image instead depicted a child with an unrelated toy-like object, since the model has no access to the real product image and generates a plausible new image purely from the text title. Specific branding, packaging text, and exact product shape are not preserved, a known limitation of text-to-image diffusion models for e-commerce use cases.

Where generation adds value: such images are best suited for quickly producing placeholder hero/lifestyle imagery for new listings before professional photography is available, or for generating marketing concept variations — not for literal, accurate product depictions.

8. Milestone 7 — Integration & Demo

All components from Milestones 1–6 were combined into a single Smart Product Assistant, wrapped in a Gradio interface. Given a selected product, the system displays:

- The actual average rating alongside the predicted rating from the Milestone 1 MLP.
- An image-based “good/not good” quality prediction from the Milestone 2 transfer-learning model.
- A list of similar products via Milestone 3 TF-IDF semantic search.
- A pros/cons review summary and a grounded question-answering response from the Milestone 5 RAG pipeline.
- An optional, on-demand generated hero image from the Milestone 6 Stable Diffusion pipeline.

The integrated demo was tested end-to-end in Google Colab on a T4 GPU and ran successfully, producing a complete report (predicted rating, image quality label, similar products, pros/cons summary, grounded Q&A answer, and a generated hero image) for a selected product without errors. Per the project guidelines, this milestone assembled the outputs of earlier milestones rather than introducing new models.

9. Conclusion & Key Takeaways

- Deep learning models improved over simple baselines in Milestones 1 (MLP > Linear Regression), 3 (sentiment classification, comparable to baseline), and 4 (DistilBERT > both M3 models).
- Milestone 2 produced an honest negative result for transfer learning on this small, domain-specific dataset: a small CNN trained from scratch outperformed a frozen MobileNetV2 backbone on macro-F1, illustrating that generic ImageNet features are not always beneficial for narrow, non-standard prediction targets with limited data.
- Macro-F1 was used throughout instead of, or alongside, raw accuracy due to strong class imbalance in both ratings and review sentiment (the large majority of reviews and products are rated positively).
- All train/validation/test splits were performed at the product level to prevent data leakage between reviews of the same product appearing in both training and evaluation sets.
- Small open-weight models (DistilBERT for classification, Flan-T5-small for generation) combined with TF-IDF retrieval were sufficient to demonstrate the full transformer and LLM/RAG pipeline within a free GPU compute budget, though the smallest generative model (80M parameters) showed clear limitations in open-ended summarization and numeric reasoning even after fine-tuning.
- The final integrated Gradio demo successfully combines all six predictive/generative components into a single, working Smart Product Assistant, fulfilling the Milestone 7 integration requirement.